

YUCHEN YANG

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PROFESSIONAL EXPERIENCES

Assistant Professor The Pennsylvania State University , PA, United States	2025.07 - Current
Research Assistant for Dr. Yinzhi Cao The Johns Hopkins University , MD, United States	2020.03 - 2025.05
Student Associate for Dr. Shao-yuan Lo , Dr. Kwonjoon Lee , and Dr. Behzad Dariush Honda Research Institute , CA, United States	2023.10 - 2024.02
Research Assistant for Dr. Yuqing Sun Shandong Univeristy , Shandong, China	2018.12 - 2019.06
Research Assistant for Dr. Yingjie Tian Chinese Academy of Sciences , Beijing, China	2018.06 - 2018.09

EDUCATION

Ph.D. The Johns Hopkins University <i>Computer Science</i> Advisor: Dr. Yinzhi Cao	2021 – 2025 MD, United States
M.S. The Johns Hopkins University <i>Security Informatics</i> Advisor: Dr. Yinzhi Cao	2019 – 2021 MD, United States
B.E. Shandong University <i>Software Engineering</i>	2015 – 2019 Shandong, China

INTERESTS

Trustworthy AI, Computer Security

PUBLICATIONS

PEER-REVIEWED CONFERENCE PAPERS (* INDICATES CO-FIRST AUTHORS)

- 2025 **CertPHash: Towards Certified Perceptual Hashing via Robust Training**
Yuchen Yang, Qichang Liu, Christopher Brix, Huan Zhang, Yinzhi Cao
In the proceedings of USENIX Security Symposium (Usenix), 2025
- 2024 **Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Behzad Dariush, Yinzhi Cao, Shao-Yuan Lo
In the Proceedings of the European Conference on Computer Vision (ECCV), 2024
- SafeGen: Mitigating Sexually Explicit Content Generation in Text-to-Image Models**
Xinfeng Li*, **Yuchen Yang***, Jiangyi Deng*, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyuan Xu
In the Proceedings of the ACM Conference on Computer and Communications Security (CCS), 2024
- RippleCOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning**

Zihao Zhao, **Yuchen Yang**, Yijiang Li, Yinzhi Cao

In the Findings of the Association for Computational Linguistics: Empirical Methods in Natural Language Processing (EMNLP), 2024

The first author finished the paper mainly under my mentoring

SneakyPrompt: Jailbreaking Text-to-image Generative Models

Yuchen Yang, Bo Hui, Haolin Yuan, Neil Gong, Yinzhi Cao

In the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2024

Media Coverage (Selected): MIT Technology Review [🔗](#)

2023 **PrivateFL: Accurate, Differentially Private Federated Learning via Personalized Data Transformation**

Yuchen Yang*, Bo Hui*, Haolin Yuan*, Neil Gong, Yinzhi Cao

In the proceedings of USENIX Security Symposium (Usenix), 2023

Earned all Artifact Badges: Artifacts Available, Artifacts Functional, Results Reproduced

Fortifying Federated Learning against Membership Inference Attacks via Client-level Input Perturbation

Yuchen Yang, Haolin Yuan, Bo Hui, Neil Gong, Neil Fendley, Philippe Burlina, Yinzhi Cao.

In the proceedings of IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023

2022 **Addressing Heterogeneity in Federated Learning via Distributional Transformation**

Haolin Yuan*, Bo Hui*, **Yuchen Yang***, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of the European Conference on Computer Vision (ECCV), 2022

2021 **Practical Blind Membership Inference Attack via Differential Comparisons**

Bo Hui*, **Yuchen Yang***, Haolin Yuan*, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of Network & Distributed System Security Symposium (NDSS), 2021

PREPRINTS

2025 **Jailbreaking Safeguarded Text-to-Image Models via Large Language Models**

Zhengyuan Jiang, Yuepeng Hu, **Yuchen Yang**, Yinzhi Cao, Neil Gong

Under review, 2025

Pseudo-Probability Unlearning: Towards Efficient and Privacy-Preserving Machine Unlearning

Zihao Zhao, Yijiang Li, **Yuchen Yang**, Wenqing Zhang, Nuno Vasconcelos, Yinzhi Cao

Under review, 2025

PROFESSIONAL SERVICES

CONFERENCE/JOURNAL REVIEWING

PC , USENIX Security Symposium	2026
IEEE Symposium on Security and Privacy (S&P)	2026
ACM Conference on Computer and Communications Security (CCS)	2025
ACM Workshop on Adaptive and Autonomous Cyber Defense (AACD)	2024
AEC , IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)	2024
Reviewer , International Conference on Learning Representations (ICLR)	2025
International Conference on Computer Vision (ICCV)	2025
IEEE Transactions on Dependable and Secure Computing (TDSC)	2023, 2024
IEEE Transactions on Information Forensics & Security (T-IFS)	2024
External Reviewer , IEEE Symposium on Security and Privacy (S&P)	2025
USENIX Security Symposium	2023, 2024
ACM Conference on Computer and Communications Security (CCS)	2022
ACM ASIA Conference on Computer and Communications Security (ASIACCS)	2024
IEEE Computer Security Foundations Symposium (CSF)	2022, 2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2022

ORGANIZING AND CHAIRING

Session Chair, IEEE Workshop on Deep Learning Security and Privacy (DLSP)

2024

ADVISING EXPERIENCE

PH.D. STUDENTS

Yichi Zhang, the Pennsylvania State University, advisor 2025.08 - current
Zhiqi Wang, the Pennsylvania State University, advisor 2025.08 - current

MASTER STUDENTS

Adila Abudurehman, master student at JHU. Now: Software engineer at Microsoft 2023.10 - 2024.02
Manoj Valeti, master student at JHU. Now: Software engineer at AWS Amazon 2023.10 - 2024.02
Amodini Vardhan, master student at JHU. Now: Security engineer at Twilio 2023.10 - 2024.02
Ashi Garg, master student at JHU. Now: Research associate at HLTCOE 2022.09 - 2023.03

UNDERGRADUATE STUDENTS

Zihao Zhao, undergraduate student at JHU. Now Ph.D. student at JHU 2023.12 - 2024.09
Qichang Liu, undergraduate student at Tsinghua University. Now Ph.D. student at UIUC 2024.06 - 2024.09
Yichen Li, undergraduate student at Beijing University 2024.06 - 2024.09
Anning Li, undergraduate student at UESTC. Now: Master student at CMU 2023.06 - 2023.09

TEACHING EXPERIENCE

Instructor, IST866: Threats and Cybersecurity, the Pennsylvania State University Fall 2025
Teaching Assistant, EN 601.340/440/640: Web Security, the Johns Hopkins University Fall 2020, 2022

MEDIA COVERAGE

Text-to-image AI models can be tricked into generating disturbing images, [MIT Technology Review](#) 2023.11
AI art generators can be fooled into making NSFW images, [IEEE Spectrum](#) 2023.11
Nonsense prompts trick AIs into producing NSFW images, [Technology Networks](#) 2023.11
Researchers reveal vulnerabilities in AI models, prompting concerns, [Cryptopolitan](#) 2023.11
AI generators can be tricked into making NSFW content, [JHU Engineering School News](#) 2023.11
SneakyPrompt: Revealing the vulnerabilities of text-to-image AI, [The Johns Hopkins News-Letter](#) 2023.12

AWARDS AND HONORS

Selected as ICLR Notable Reviewer 2025.05
Selected to attend Individualized Cybersecurity Research Mentoring Workshop (iMentor) 2021.11
Academic Scholarship, Shandong University 2018.09

PATENT

2024 **System and Method Using Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Shao-Yuan Lo, Behzad Dariush
Under US Patent Application (Filed), 2024

2018 **A New System for Stock Volatility Prediction by Using Privileged Support Vector Machines**
Yuchen Yang, Zheng Yan, Haoyang Li, Zhixuan Lv, Weiwei Zhao, Simiao Zhao
Under IP Australia Application (No.2018101304), 2018

INVITED TALKS

"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", The Pennsylvania State University, Invited talk	2025.3
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Case Western Reserve University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Rice University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Virginia Tech, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Rochester Institute of Technology, Invited talk	2025.1
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Monash University, Invited talk	2024.10
"Zero-shot Video Anomaly Detection: Steering LLMs for Rules-based Reasoning", Voxel51, Invited talk	2024.11